



**ICPC2T
2027**



Student Branch
NIT Raipur



2027 IEEE International Conference on

Power, Control and Computing Technologies (ICPC2T 2027)

NIT Raipur, Chhattisgarh, India | 3–5 March 2027

SPECIAL SESSION (SS10) ON

Synergising the Hydrogen Transition, Electric Mobility, and Power Grid Resiliency for Global Sustainability

LIST OF TOPICS

- Machine Learning Algorithms for Predictive Energy Management in Hydrogen-EV-Grid Ecosystems
- Evaluating Grid Resilience: The Role of Green Hydrogen in Mitigating EV Demand Peaks
- Thermodynamic Modelling of Solid Oxide Fuel Cells for Combined Heat, Power, and Hydrogen (CHPH) Grids
- Grid-to-Gas (G2G) Architectures: Utilizing Excess EV Grid Capacity for Hydrogen Electrolysis
- Dynamic Modelling of Power Grids under High Penetration of Hydrogen Generation and EV Charging
- Hydrogen Energy Storage Systems (HESS) for Long-Duration Grid Stabilization
- Smart Inverter Topologies for Integrating Fuel Cells into Renewable Micro grids
- Techno-Economic Analysis of Hydrogen Fuel Cell Electric Vehicles (FCEVs) vs. Battery Electric Vehicles (BEVs)
- Lifecycle Assessment of Hydrogen Production Pathways for Clean Transportation
- Developing Hybrid Hydrogen-Battery Storage Systems for Heavy-Duty Electric Vehicles
- Safety Protocols and Materials Science in High-Pressure Hydrogen Refuelling Infrastructure

ORGANIZING CHAIRS



Dr. Sonali Paunikar

Sarvepalli Radhakrishnan
University, Bhopal



Dr. N. P. Patidar

MANIT, Bhopal



Dr. M. L. Kolhe

University of Agder,
Norway



Dr. Balakrushna Sahu

Hitachi Energy, IDC,
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After login, select Special Session (SS10): **“Synergising the Hydrogen Transition, Electric Mobility, and Power Grid Resiliency for Global Sustainability”**